

Proposal for NETSOL AI Hackathon 2024

1. Team Information

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2. Problem Statement

We selected Problem Number 1: Predictive Maintenance for Leased Vehicles.

3. System Overview

Our AI-based Predictive Maintenance System (PMS) for leased vehicles helps the leasing company keep their cars in good shape. By predicting and addressing potential issues before they occur, our system ensures timely maintenance, reduces unexpected breakdowns, and enhances customer satisfaction.

The model will use AI and machine learning to analyze data collected from the car's sensors and past maintenance records. By identifying patterns, it will predict possible problems and prioritize them, focusing on critical issues first. This strategy, known as Optimized Resource Allocation, makes maintenance more efficient by avoiding unnecessary repairs.

Our PMS will continuously learn and improve with new data and technology, making it more accurate over time. The result is a smart maintenance system that reduces vehicle downtime, lowers costs, and increases vehicle performance.

4. Technical Approach

Problem

Leased vehicles often face sudden breakdowns, causing inconvenience to customers, higher maintenance costs for companies, and damage to the vehicle's reputation. Traditional maintenance methods, which fix problems only after they occur or at scheduled times, are costly and inefficient.

Our Solution

We propose an AI-powered Predictive Maintenance System that uses machine learning to analyze sensor data and predict vehicle failures. Here's how it works:

1. Data Collection and Preprocessing

- **Sensor Data:** Collect real-time information from the car's components, like the engine, brakes, and tire pressure.
- **Historical Records:** Use past repair and maintenance records.
- **External Data:** Include information like weather and traffic conditions.
- **Data Cleaning:** Process and clean the data to remove errors or gaps.

2. Feature Engineering

- **Extract Key Features:** Identify important data points like speed, temperature, and sensor readings.
- **Time-Series Features:** Analyze trends over time to identify patterns, using averages and variations.

3. Model Development and Training

- **Deep Learning Models:** Use advanced models like RNNs, LSTMs, and CNNs to analyze time-series data and spot anomalies.
- **Ensemble Learning:** Combine several models to increase accuracy.
- **Transfer Learning:** Use pre-trained models to speed up training, especially when data is limited.

4. Model Deployment and Monitoring

- **Real-Time Predictions:** Use the trained models to make instant predictions from new sensor data.
- **Alert System:** Notify the maintenance team when an issue is detected and suggest actions.
- **Model Monitoring:** Regularly check model performance and update as needed.

Innovation and Impact

Our system is unique because it combines multiple advanced techniques and focuses on proactive maintenance. By predicting problems early, we can reduce vehicle downtime, cut maintenance costs, and improve reliability.

Impact in the Real World:

- **Lower Costs:** Less downtime and smarter maintenance scheduling save money.
 - **Higher Customer Satisfaction:** Fewer breakdowns mean happier customers.
 - **Longer Vehicle Life:** Quick and timely repairs help vehicles last longer.
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Challenges and Solutions

Potential Challenges:

- **Data Quality:** We need accurate and enough data to make reliable predictions.
- **Model Complexity:** Building and running complex models can be difficult.
- **Real-Time Processing:** Handling large amounts of data quickly is a challenge.

Solutions:

- **Data Cleaning:** Implement thorough data cleaning methods.
- **Step-by-Step Model Building:** Start simple and gradually increase complexity.
- **Cloud Deployment:** Use cloud services to handle real-time data efficiently.
- **Continuous Improvement:** Keep monitoring and improving the system.

By addressing these challenges, our AI-based PMS will transform vehicle maintenance, save costs, and boost customer satisfaction and vehicle performance